70 Ophiuchi AB: Hypothetical Planet Life-Prospect Screening

Equilibrium-climate and eccentricity stress test performed October 10, 2026. Flux habitability limits (0.2-1.1 S_earth) follow the numerical study of 70 Oph / 36 Oph / gamma Leo (arXiv:2608.13243). Stellar parameters: 70 Oph A (0.88 Msun, 0.53 Lsun) and 70 Oph B (0.75 Msun, 0.15 Lsun), binary separation ~23 AU, e=0.5. This is a simplified analytic screening, not a full N-body integration.

1. Habitable-zone inventory (conservative recent-Venus / early-Mars limits)

Star	HZ width (AU)	T_eq range (K)	Year length (days)	Notes
70 Oph A	0.75-1.25	195-251	253-544	Earth-like; no tidal locking
70 Oph B	0.39-0.64	198-254	103-216	Likely tidally locked; 'eyeball' climate regime

2. Binary eccentricity stress test (Kozai-Lidov cycles)

Star / orbit	Mutual inclination	e_max	Time in habitable flux	Verdict
70 Oph A @ 0.93 AU	0 deg	0.0	100.0%	Safe
70 Oph A @ 0.93 AU	30 deg	0.0	100.0%	Safe
70 Oph A @ 0.93 AU	45 deg	0.408	90.8%	Marginal - periodic climate excursions
70 Oph A @ 0.93 AU	60 deg	0.764	84.6%	Stressed - long climatic winters/summers
70 Oph B @ 0.45 AU	0 deg	0.0	100.0%	Safe
70 Oph B @ 0.45 AU	30 deg	0.0	100.0%	Safe
70 Oph B @ 0.45 AU	45 deg	0.408	84.7%	Stressed - long climatic winters/summers
70 Oph B @ 0.45 AU	60 deg	0.764	81.9%	Stressed - long climatic winters/summers

3. Conclusions

- Planets coplanar with the binary (i < 39.2 deg, the critical Kozai angle) experience no eccentricity cycling and remain in habitable flux 100% of the time - consistent with published simulations showing only 1.5-1.8% becoming uninhabitable.

- Misaligned worlds (45 deg) spend 85-91% of time in habitable flux; orbit-averaged flux stays habitable, but deep periastron summers (up to ~2.7x Earth flux at e=0.41) could stress fixed-atmosphere climates. Oceans/ice provide buffering (Earth analog: e=0.0167, Milankovitch cycles).

- No giant planets exist in either HZ, which removes a major scattering threat and means any terrestrial world that formed has had ~billions of years of quiet evolution.

- Caveats: planet formation in 23-AU binaries is less efficient; stellar activity (the same signal that mimicked planets for a century) could erode atmospheres; no planets have actually been detected yet.